

$$\phi: V \longrightarrow \{1, 2, \dots, k\}$$

$$\phi(u) \neq \phi(v) \quad \forall u, v \in E$$

$$V = V_1 \cup V_2 \cup \dots \cup V_k \quad \text{such that}$$

$$V_i = \phi^{-1}(i)$$



$$\sqrt{\mathcal{E}}(G, \phi) = (\sqrt{V}, \mathcal{E}) \text{ of}$$

$$G = (V, E)$$



$$v_\ell(\langle u, v \rangle) \Rightarrow$$

A tree diagram showing a central node labeled $\langle u, v \rangle$ connected to three other nodes: $\langle x, z \rangle$, $\langle w, z \rangle$, and $\langle z, b \rangle$. The edges are labeled with blue '+' and '-' signs.



— 3-Cycle Problem —

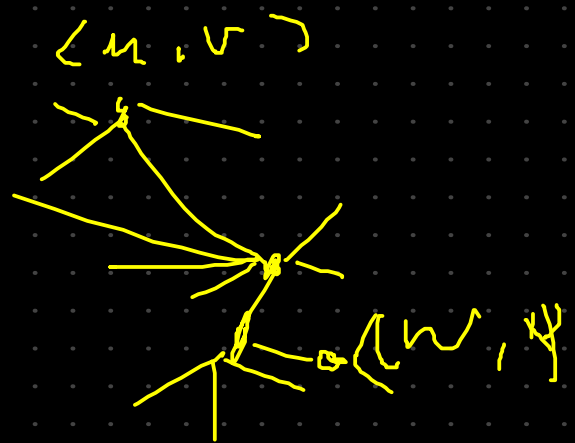
$u, v, w \in V$

$u < v < w < u$

$u, \mathcal{L} \sqrt{\mathcal{E}}(u, \phi)$

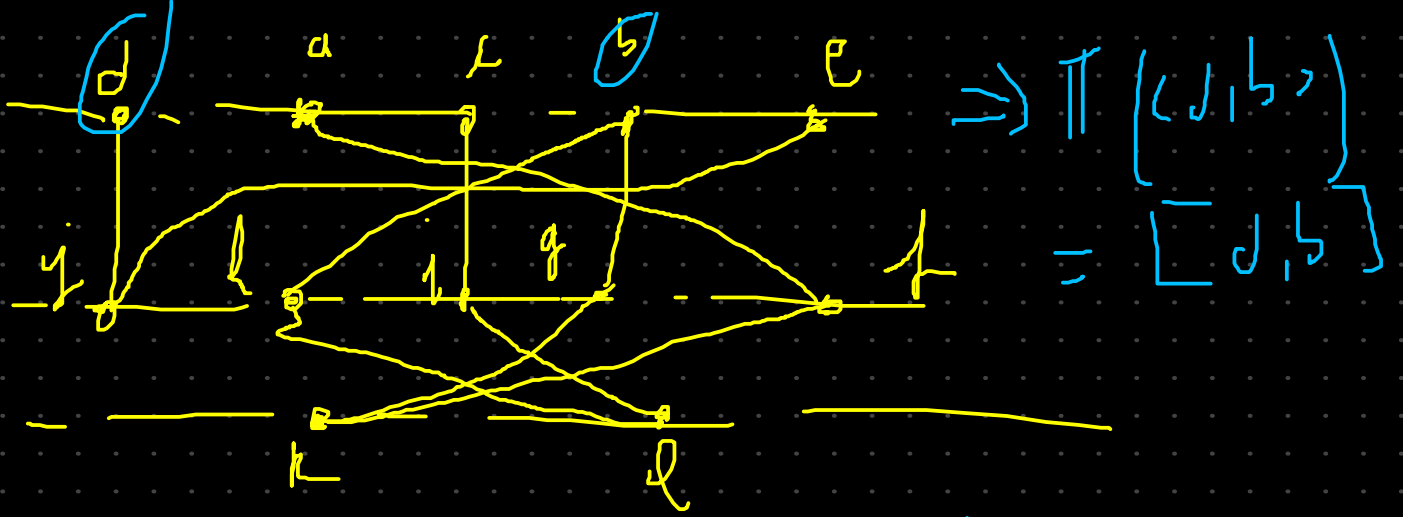
$\langle e, \beta \rangle, \langle \beta, \gamma \rangle, \langle \gamma, e \rangle$

$\mathcal{L}$ construct
 $\xrightarrow{\text{Mapping } \pi}$



$\pi V \rightarrow V$

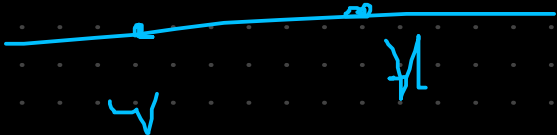
$\pi(\langle u, v \rangle) = [v, u]$
 $\pi(\langle w, y \rangle) = [y, w]$

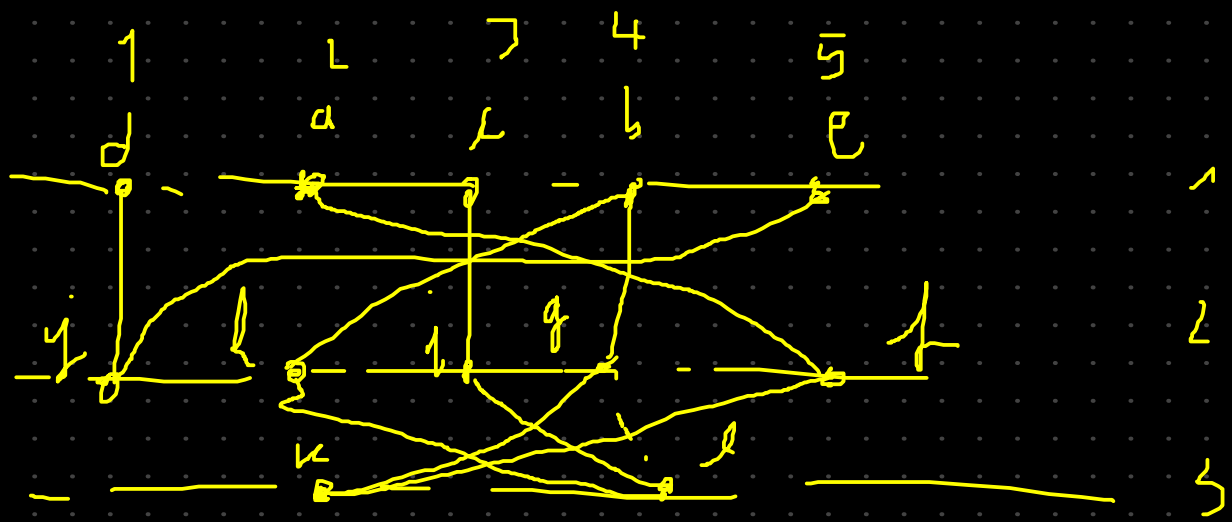


$$S = \left(\langle d, b \rangle \right)$$



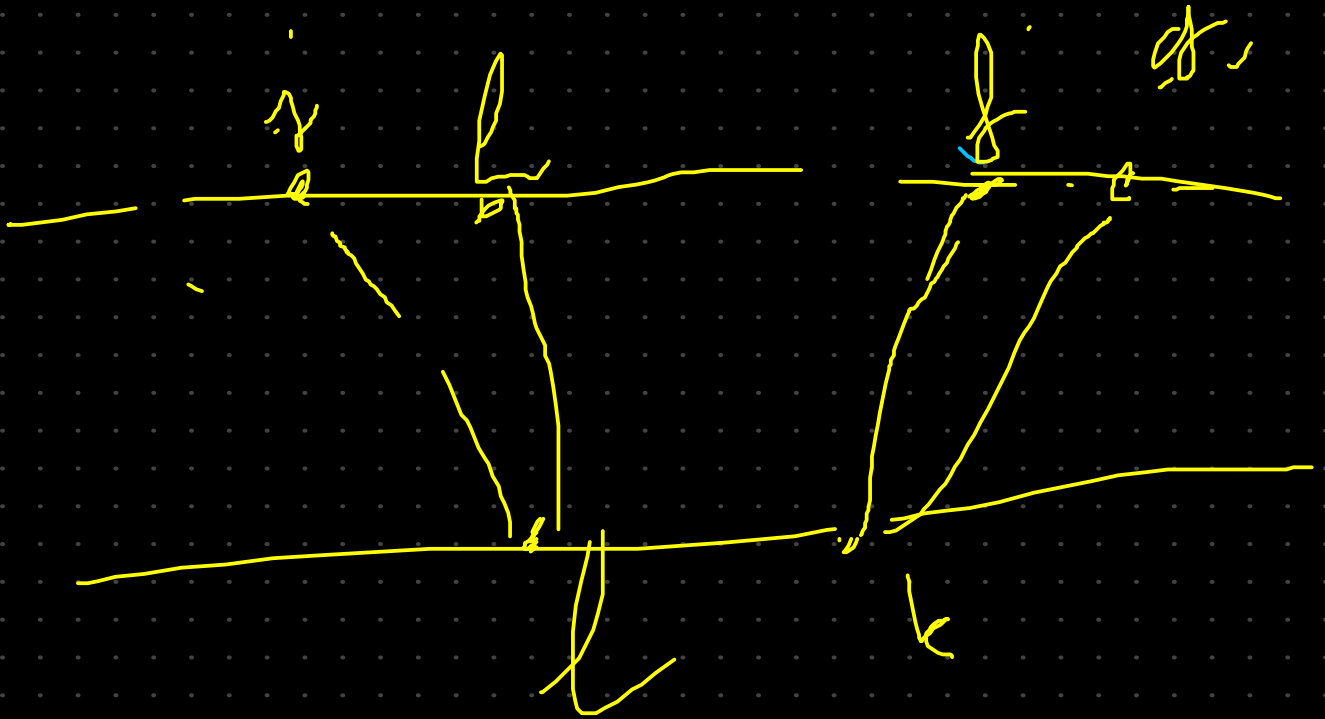
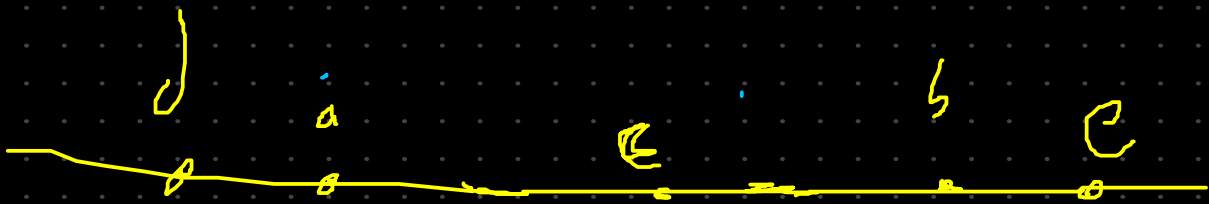
$$\Pi \left(\langle w, v \rangle \right) \neq \begin{cases} [w, v] & \lambda = + \\ [v, w] & \lambda = - \end{cases}$$





$$\langle u, v \rangle = \text{vertexAt}(\mathcal{G}, i, j)$$

$$\langle u, v \rangle \mid \begin{array}{l} \text{pluscnt} = 1 \\ \text{minuscnt} = 1 \end{array}$$



$$S = \left(\begin{array}{c} \langle d, c \rangle, \langle d, e \rangle, \langle a, b \rangle \\ \langle a, e \rangle, \langle j, g \rangle, \langle b, c \rangle \\ \text{true} \end{array} \right)$$

